American Computer Science League

2021-2022 ● Contest 1: Short Problems ● Elementary Division

1. Computer Number Systems

Convert 147_8 to base 10.

A. 481
B. 120
C. 114
D. 103
E. 93

2. Computer Number Systems

Which of the following has the fewest number of 1s in its binary representation?

$9A7_{16}$ $B14_{16}$ 7431_8 4265_8

A. 9A7
B. B14
C. 7431
D. 4265
E. more than 1 answer

3. Computer Number Systems

If this year is 84A in hexadecimal, what would the hexadecimal year be 3F7 hexadecimal years from now?

A. C40
B. B31
C. D42
D. 1241
E. C41

4. Computer Number Systems

Determine which of the following is the largest number:

111001101010_2 $E60_{16}$ $E69_{16}$ 7147_8 7150_8

A. 111001101010
B. E60
C. E69
D. 7147
E. 7150

5. Computer Number Systems

What is the base 10 equivalent to the most dominant (the one with the largest value) component in the hexadecimal color "#9FA0B1"?

A. 193
B. 177
C. 176
D. 160
E. 159

美国计算机科学联赛

2021-2022 ● 第一轮: 短问题 ● 小学组

1. 计算机数字系统

请将 147_8 转化为十进制。

A. 481
B. 120
C. 114
D. 103
E. 93

2. 计算机数字系统

将下列数字转化成二进制后，哪一个含有的 1 最少?

$9A7_{16}$ $B14_{16}$ 7431_8 4265_8

A. 9A7
B. B14
C. 7431
D. 4265
E. 答案不止一个

3. 计算机数字系统

若今年是 $84A_{16}$，那么从现在起 $3F7_{16}$ 年后以十六进制表示会是哪一年？

A. C40
B. B31
C. D42
D. 1241
E. C41

4. 计算机数字系统

下列数字中最大的是哪个？

111001101010_2 $E60_{16}$ $E69_{16}$ 7147_8 7150_8

A. 111001101010
B. E60
C. E69
D. 7147
E. 7150

5. 计算机数字系统

下列十进制的数中，哪一个和十六进制颜色 “#9FA0B1” 中最大的一部分（值最大）相等？

A. 193
B. 177
C. 176
D. 160
E. 159